

①

$$A^4 = P D^4 P^{-1}$$

$$P^{-1} = \begin{bmatrix} 3 & -1 \\ -2 & 5 \end{bmatrix}$$

$$D^4 = \begin{bmatrix} 16 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 5 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 16 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 226 & -523 \\ 80 & -208 \end{bmatrix}$$

③

$$A = P D P^{-1} \quad A^k = ?$$

$$\begin{bmatrix} 2 & 0 \\ 3(26) & 6 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$$

$$A^k = P D^k P^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2^k & 0 \\ 0 & 6^k \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} 2^k & 0 \\ 3(2^k - 6^k) & 6^k \end{bmatrix}$$

⑤

$$\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & -\frac{3}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$$

$$\lambda = 5, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \lambda = 1: \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$$

eigenvalues: 5, 1

④

$$\begin{bmatrix} 1 & 0 \\ 6 & -1 \end{bmatrix}$$

$$\lambda = \pm 1$$

$$\lambda = -1$$

$$A - I = \begin{bmatrix} 0 & 0 \\ 6 & -2 \end{bmatrix}$$

$$(A - I)x = 0, \quad 6x_1 - 2x_2 = 0 \Rightarrow x_1 = \frac{1}{3}x_2$$

x_2 is free

$$x_2 = \begin{bmatrix} \frac{1}{3} \\ 1 \end{bmatrix} \text{ basis vector for eigenspace } v_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\lambda = -1$$

$$A + I = \begin{bmatrix} 2 & 0 \\ 6 & 0 \end{bmatrix}$$

$$(A + I)x = 0: 2x_1 = 0 \Rightarrow x_1 = 0, x_2 \text{ is free}$$

$$x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ basis for eigenspace } v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$P = [v_1, v_2] = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$⑧ \begin{bmatrix} 3 & -1 \\ 1 & 5 \end{bmatrix}$$

$$\det(A - \lambda I) = \det \begin{bmatrix} 3-\lambda & 1 \\ 1 & 5-\lambda \end{bmatrix} = \lambda^2 - 8\lambda + 16 =$$

$$= (\lambda - 4)^2 \Rightarrow \lambda = 4$$

$$A - 4I = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$(A - 4I)x = 0, \quad x_1 + x_2 = 0 \Rightarrow x_2 \text{ is free}$$

$$x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

no λ , A is not diagonalizable

⑨

$$\lambda = 3, \quad A - 3I =$$

$$\begin{bmatrix} -4 & 4 & -2 \\ -3 & 1 & 0 \\ -3 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -\frac{1}{4} & 0 \\ 0 & 1 & -\frac{3}{4} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_3 \begin{bmatrix} 1 \\ \frac{3}{4} \\ \frac{1}{4} \end{bmatrix} \Rightarrow \text{basis: } v_1 = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$$

$\lambda = 2$,

$$A - 2I =$$

$$\begin{bmatrix} -3 & 4 & -2 \\ -3 & 2 & 0 \\ -3 & 6 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -\frac{2}{3} & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_3 \begin{bmatrix} \frac{2}{3} \\ 1 \\ 1 \end{bmatrix}$$

$\Rightarrow$ basis: $v_2 =$

$$\begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}$$

$\lambda = 1$,

$$A - I =$$

$$\begin{bmatrix} -2 & 4 & -2 \\ -3 & 3 & 0 \\ -3 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$\Rightarrow$ basis: $v_3 =$

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$P =$

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 3 & 1 \\ 4 & 3 & 1 \end{bmatrix}$$

$D =$

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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$$\lambda = 5:$$

$$A - 5I = \begin{bmatrix} -3 & 2 & -1 \\ 1 & -2 & -1 \\ -1 & -2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 10 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_3 \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} \Rightarrow \text{basis } v_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

$$\lambda = 1:$$

$$A - I = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & -1 \\ -1 & -2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \{v_1, v_3\} = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$P = \begin{bmatrix} -1 & -2 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(15)

$$\lambda = 3:$$

$$A - 3I = \begin{bmatrix} 4 & 4 & 16 \\ 2 & 2 & 8 \\ -2 & -2 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \{v_1, v_2\} = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\lambda = 1:$$

$$A - I = \begin{bmatrix} 6 & 4 & 16 \\ 2 & 4 & 8 \\ -2 & -2 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_3 \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix} \Rightarrow v_3 = \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}$$

$$P = \begin{bmatrix} -1 & -4 & -2 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(1)

$$\begin{bmatrix} 4 & 0 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$\lambda = 4, 5$$

$$\lambda = 4:$$

$$A - 4I = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$v_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$\lambda = 5 \Rightarrow A$ is not diagonal $\text{eig } \lambda > 1$

(2)

$$\begin{bmatrix} 5 & -3 & 0 & 8 \\ 0 & 3 & 1 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$\lambda = 2, 3, 3$$

$$\lambda = 2:$$

$$A - 2I = \begin{bmatrix} 3 & -3 & 0 & 8 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \{v_1, v_2\}$$

$$\begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$$\lambda = 3:$$

$$A - 3I = \begin{bmatrix} -2 & -3 & 0 & 8 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -\frac{3}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_2 \begin{bmatrix} 3 \\ 2 \\ 1 \\ 0 \end{bmatrix} \Rightarrow v_3 = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda = 5:$$

$$A - 5I = \begin{bmatrix} 0 & -3 & 0 & 8 \\ 0 & -2 & 1 & -2 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & -3 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow v_4 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$D = \begin{bmatrix} -1 & -1 & 3 & 1 \\ -1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$